

Semester Project

Predicting Cross-Border Transmission
Capacities in the European Electricity
Grid using different Machine Learning
Architectures

Andrej Bernhard
August 22, 2025

Advisors

Prof. Dr. Florian Dörfler
Dr. Saverio Bolognani
Dr. Keith Moffat
Fabian Gottschlich (BKW)
Ronny Schneeberger (BKW)

Abstract

This project addresses the challenge of predicting cross-border transmission capacities in the European electricity grid, a critical input for the long-term price prognosis (LPP) model of BKW. A machine learning model was developed using publicly available data for training to predict border capacities in two capacity allocation domains: Net Transfer Capacity (NTC) and Flow-Based Market Coupling (FBMC).

Four different model architectures were compared, including a simplistic base model to benchmark on, a fully connected neural network, a long short-term memory (LSTM) model, and a novel Hybrid model featuring residual connections and dynamic selection between classification and regression tasks depending on border characteristics. Feature engineering involved a tailored feature selection method based on geographic adjacency, principal component analysis, and manual normalization of cross-border capacities.

Comparison between the models showed that the Hybrid model outperforms the others, likely due to the implementation of residual connections. Overall, FBMC borders yielded consistently more accurate predictions, presumably due to more standardized allocation mechanisms, while certain NTC borders remained more difficult to predict precisely.

To assess practical value, BKW conducted a backtest by feeding the model's outputs, alongside their standard capacity profiles and historical ground truth values into the operating LPP-model. This integration led to a reduction in the mean absolute error of electricity price forecasts by up to 20.5% across bidding zones, approaching the theoretical limit achieved with perfect capacity information. These results highlight the practical use of machine learning to improve electricity price forecasts through enhanced cross-border capacity modeling.

Preface

Conducting a case study in collaboration with BKW AG in the first year of my academic journey in the MEST (Master of Science in Energy Science and Technology) program at ETH Zurich was the starting point of this collaboration. I first got to know the team from BKW during this case study, where we developed a model to compare solar electricity production and residential cooling demand in Europe. This sparked my interest in modeling demand and supply in the energy sector and creating code-based solutions to real-world energy challenges.

Motivated by that experience, I reached out to BKW with the intention of continuing our collaboration. They explained to me the problem of estimating and predicting cross-border transmission capacities in the European power system. The idea was to build a machine learning model which can learn on historical data and can then be applied to make predictions for the future. In addition to the model developed in the case study, this project should also be part of the long-term electricity price forecasting strategy.

I would like to thank Fabian Gottschlich and Ronny Schneeberger from the BKW Strategic Market Analysis team for their support and guidance throughout this project. Taking the time to discuss questions, together with their insights, made this project both feasible and rewarding. I am also grateful to Saverio Bolognani and Keith Moffat from the IfA Laboratory at ETH Zurich for their support during the project. Their valuable inputs and help with organizational matters were much appreciated.

Contents

List of Figures	ix
List of Tables	xi
1 Introduction	1
1.1 Electricity Long-term Price Prognosis Model	1
1.2 Types of Cross-Border Capacity Allocation	2
1.3 Related Work	3
1.3.1 Artificial Intelligence in the Energy Market	3
2 Methods	5
2.1 Data Preprocessing	5
2.1.1 Hypothesis Verification	5
2.1.2 Feature Engineering	7
2.1.3 Column Filtering and PCA	8
2.1.4 Normalization	9
2.2 Modeling	10
2.2.1 Regression Models	11
2.2.2 Hybrid Model	12
3 Results	15
3.1 Training Conditions	15
3.2 FBMC Results	16
3.3 NTC Results	19
3.4 Detailed Hybrid Model Analysis	21
3.4.1 Backtest	22
4 Discussion	25
4.1 Problems	25
4.1.1 Real-World Events	25
4.1.2 Feature Columns	25
4.1.3 Border Division	26
4.2 Conclusion	26
4.2.1 Baseline and Feature Extended Dataset	26

4.2.2	FBMC and NTC Borders	26
4.3	Outlook	27
4.3.1	Modeling Improvements	27
4.3.2	Data Expansion and FBMC Focus	28
Bibliography		28
A Appendix		31
A.1	Detailed Border Division	31
A.2	GitHub Repository	32
A.3	Figures	32

List of Figures

1.1	LPP model with input assumptions and resulting outputs. ¹	1
1.2	Comparison of the NTC and FBMC capacity allocation domains. ²	2
2.1	Cross-border capacities during colder months (blue) and warmer months (yellow).	6
2.2	Aggregated MAXBEX values for three specific borders on weekdays and weekends. ¹⁰	6
2.3	Correlation map of some of Germany's export borders and onshore wind generation in Germany.	7
2.4	Feature engineering map with two resulting main datasets	8
2.5	Illustration of neighboring country filtering for two different borders.	9
2.6	Border from Austria to Italy before "max-so-far" normalization.	10
2.7	Border from Austria to Italy after "max-so-far" normalization.	10
2.8	Comparison of base model and net architecture	11
2.9	Schematic diagram of LSTM model architecture. ¹³	12
2.10	NTC border differentiation for hybrid model.	12
2.11	Hybrid model architecture with detailed view on residual block structure.	13
3.1	Average training MAE (blue) and validation MAE (orange) over time for different models on FBMC borders (full sized in the Appendix).	16
3.2	Average MAE in testing for each model on BL and FX dataset (full sized in the Appendix).	17
3.3	Per border R^2 -scores for each model on BL dataset.	17
3.4	Germany to Austria border with training, validation, and test split.	18
3.5	Predictions of every model on the border from Germany to Austria.	18
3.6	Average training MAE (blue) and validation MAE (orange) over time for different models on NTC borders (full sized in the Appendix).	19
3.7	Average MAE in testing for each model on BL and FX dataset.	20
3.8	Per border MAE scores for each model on BL dataset.	20
3.9	Predictions of every model on the border from Italy to Switzerland.	21
3.10	Overview of MAE on all borders for hybrid model (full sized in the Appendix).	22
3.11	Schematic flowchart of backtest process performed by BKW.	22

3.12	Comparison of MAE across models, showing improvement over BKW's standard profiles.	23
4.1	Schematic overview of temporal convolutional layer without dilation . . .	27
A.1	Average training MAE (blue) and validation MAE (orange) over time for different models on FBMC borders.	33
A.2	Average MAE in testing for each model on BL and FX dataset.	34
A.3	Average training MAE (blue) and validation MAE (orange) over time for different models on NTC borders.	35
A.4	Average MAE in testing for each model on BL and FX dataset.	36
A.5	Overview of MAE on all borders for Hybrid model.	37

List of Tables

2.1	Overview of data types and categories	5
2.2	Overview of models	11
3.1	Training hyperparameters for all models	15
A.1	Breakdown of borders into regression and classification	31

Chapter 1

Introduction

1.1 Electricity Long-term Price Prognosis Model

The Strategic Market Analysis Team at BKW AG is responsible for electricity price forecasts in Europe up to 2050, using a long-term price prognosis model (LPP) solving an optimization on the merit order curve that requires several inputs, including assumptions on renewable expansion, demand development, and especially transmission capacities between bidding zones, commonly referred to as **interconnectors**. These cross-border capacities significantly influence absolute prices and price spreads. Transmission system operators (TSO), responsible for ensuring N-1 grid stability, apply detailed grid models to determine the allowed capacities between bidding zones. The functionality of the price forecast model is shown in the figure below.

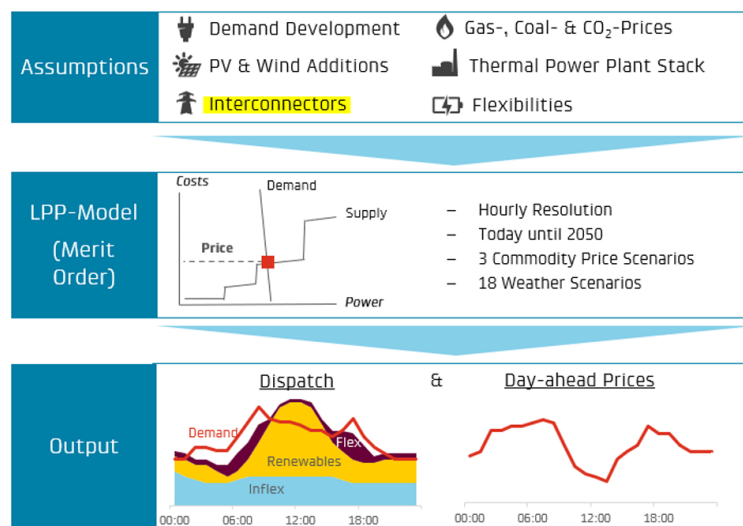


Figure 1.1: LPP model with input assumptions and resulting outputs.¹

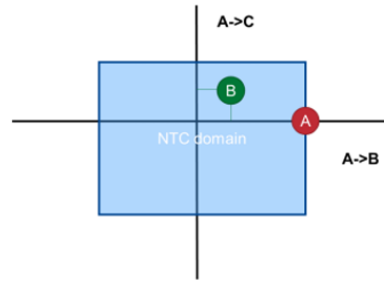
¹Provided by Strategic Market Analysis Team, BKW AG

1.2 Types of Cross-Border Capacity Allocation

In the European electricity grid, two different capacity allocation methods are used for interconnectors. Some countries and bidding zones participate in the flow-based market coupling (FBMC) area, while others rely on predefined net transfer capacities (NTC). More and more countries are joining the FBMC-Zone. For example, Nordic countries (Denmark, Finland, Norway, and Sweden) have transitioned to the FBMC-Zone in October 2024.

NTC Domain

- **NTC limits:** import and export limit for every border
- **Market result A:** Capacity in A→B direction is used fully, no Exchange in direction A→C
 - Regardless of Exchange in A→C direction, more Exchange in A→B direction is not allowed
- **Market result B:** Transaction in direction A→B and A→C



FB Domain

- **FB limits:** lines = CNEC
 - RAM – distance from origin, PTDF – incline
- **Market result A:** Capacity in A→B direction is used fully, no Exchange in direction A→C
 - More transaction in A→B direction is allowed if there is negative transaction in A→C direction

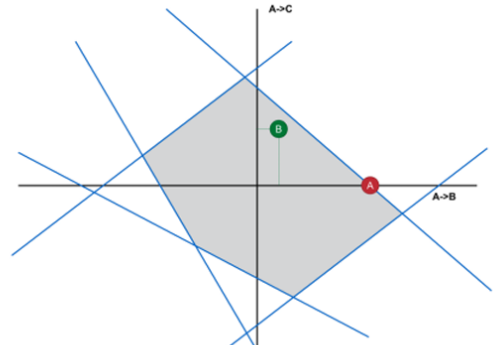


Figure 1.2: Comparison of the NTC and FBMC capacity allocation domains.²

Allocation principles

In the NTC domain, capacities are determined bilaterally by the TSOs involved in the two countries. Based on pre-defined scenarios concerning the country's electricity export and production, both TSOs suggest a capacity, and the lower one is set for the border. On the other hand, capacities in the FBMC-Zone are the result of an overall optimization in the European power grid system; therefore, being more continuous and dependent on each other, enabling potentially higher grid utilization as it more accurately reflects the physical state of the underlying network and can adapt more dynamically. In this project, the "Maximum Bilateral Exchange" (MAXBEX) is set as the target variable for FBMC borders, which corresponds to the highest possible electricity exchange between two bidding zones, given that all other net export positions are set to 0. Therefore, the MAXBEX value must be scaled relative to the actual cross-border capacity.

²<https://www.jao.eu/core-fb-da-mc>

1.3 Related Work

In 2017, two years after the introduction of flow-based market coupling, Marten Ovaere et al. [1] reported that cross-border exchange values had fallen to $400 \frac{MWh}{h}$ below pre-FBMC levels. However, when controlling for external factors (e.g. renewables, outages, fuel prices), FBMC still showed a positive long-term effect of $+1000 \frac{MWh}{h}$ on exchange volumes, reaching an estimated welfare gain of around €116 millions. This emphasizes the benefits over the older NTC method and why more and more countries are switching to the FBMC-Zone.

Another study conducted in 2011 by Goulven Salic and Yann Rebours [2] investigated the influence of wind generation forecasts on net transfer capacities (NTCs) within Germany’s electricity grid. Their research reveals an inverse relationship: As wind generation forecasts increase, the available NTCs tend to decrease with an approximate correlation of around 14% of the observed variability. An important underlying reason is the TSOs limiting the capacities to maintain grid stability. The study underscores the importance of incorporating publicly available data into predictive models for cross-border capacities.

1.3.1 Artificial Intelligence in the Energy Market

On the other hand, the worldwide importance of energy resources and price predictions and the application of machine learning in the energy market have gained strong popularity in recent years, but topics such as cross-border capacities still have to be solved. Political events, exceptional weather conditions, and human interactions are challenging topics to model and depict accurately.

In 2019 Hazem Abdel-Khalek et al. [3] reported an attempt to predict cross-border capacities in the FBMC-Zone using only publicly available data for up to 12 hours ahead using a non-linear autoregressive exogenous model (NARX). However, the low accuracy of the prediction was explained by the lack of a detailed understanding of the structure and evolution of the flow-based capacity domain and its relationship to the underlying characteristics of the market and infrastructure.

Feature Engineering Approaches

Yago Saez et al. [4] investigated a different scenario in the year 2019, in which random forests were used to select and reduce features using an iterative process to analyze congestion patterns and price convergence in Central Western Europe. This led to the conclusion that the growth of renewable energy production can interfere with the electricity market if transmission lines are not adapted to the new flow-based market coupling paradigm. Bashir Bakhshideh Zad et al. [5] introduced a novel single-step model in 2022 that directly maps demand and generation data to flow-based domains using the K-nearest neighbors algorithm, criticizing the current two-step approach used by TSOs, which involves clustering and correlation of historical flow-based domains based on simplified assumptions. This approach resulted in an increase in accuracy, scalability, and computational efficiency in modeling flow-based domains.

Probabilistic Models

Recently, probabilistic models are gaining increased attention in addition to classical machine learning architectures. In 2024 Girod et al. [6] developed a reinforcement learning agent to integrate congestion costs into cross-border energy balancing bids. This technique demonstrated a improvement in welfare and how artificial intelligence can enhance real-time capacity allocation in markets through clever congestion management. Similarly, the National Energy System Operator (NESO) publicly owned by Great Britian published the annual report [7] from 2023 where a tree-based ensemble learning model is built to estimate probability distributions of deviations between scheduled and actual interconnector flows improving, e.g. the root mean squared error over point forecasts by 14% for the interconnector capacity between GB and the Netherlands.

A promising and effective model to predict electricity prices was proposed in 2025 by He Jiang et al. [8] adaptively fusing pre-trained probability models, resulting in increased flexibility and robustness especially due to the usage of residual connections of pre-trained models such as LassoNet or NBeatsX.

Chapter 2

Methods

2.1 Data Preprocessing

Using only publicly available data sources, the following data was gathered, which is listed in the table below.

Table 2.1: Overview of data types and categories

Type of Data	Categories
Electricity profiles ^{3,4,5}	Demand and generation profiles (wind, solar, fossil, etc.)
Weather ⁶	Temperature, wind speed, rain, solar irradiance
Country parameters ^{3,7}	Area, population
Network parameters ⁸	Network utilization, congestion counter
Cross-border capacities ^{3,4,5}	Hourly NTC and MAXBEX values
Electricity prices ^{3,9}	Hourly profiles for bidding zones

2.1.1 Hypothesis Verification

To begin, a list of hypotheses was prepared with the correlations expected to be found in the data to verify the data set. This step proved to be of importance since it shed light on gaps and unusual values in the data. The following examples show correlations expected to be found in the data from logical consequences or literature review.

³ENTSO-E Transparency Platform: <https://transparency.entsoe.eu>

⁴NordPool Group: <https://data.nordpoolgroup.com/>

⁵ELEXON BSC Insight Solutions: <https://bmrs.elexon.co.uk/>

⁶NASA MERRA-2: <https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/>

⁷Wikipedia: <https://en.wikipedia.org/>

⁸EEX Transparency: <https://www.eex-transparency.com/>

⁹JAO Publication Tool: <https://publicationtool.jao.eu/>

1. Influence of temperature: During colder months average cross-border capacities increase due to higher naturally convective cooling of transmission lines.

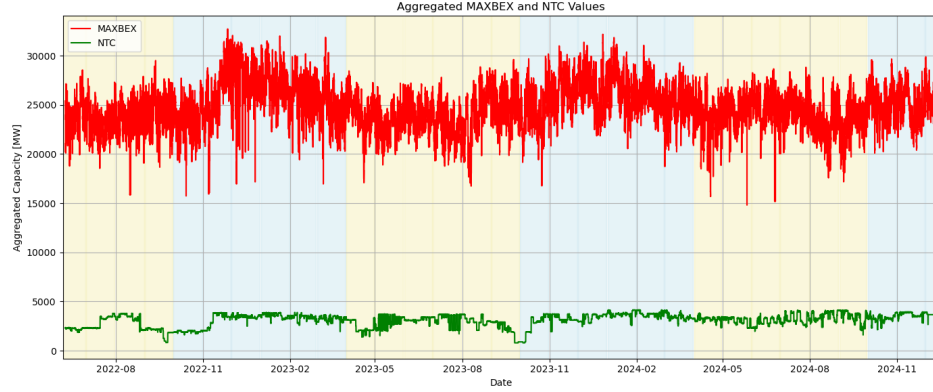


Figure 2.1: Cross-border capacities during colder months (blue) and warmer months (yellow).

2. Weekdays and weekends: Cross-border capacities are higher on weekends than on weekdays because of a lower power demand on weekends and fewer chances of critical pre-loads inside the national power systems. Therefore, TSOs can allow for higher cross-border electricity exchange between different bidding zones.

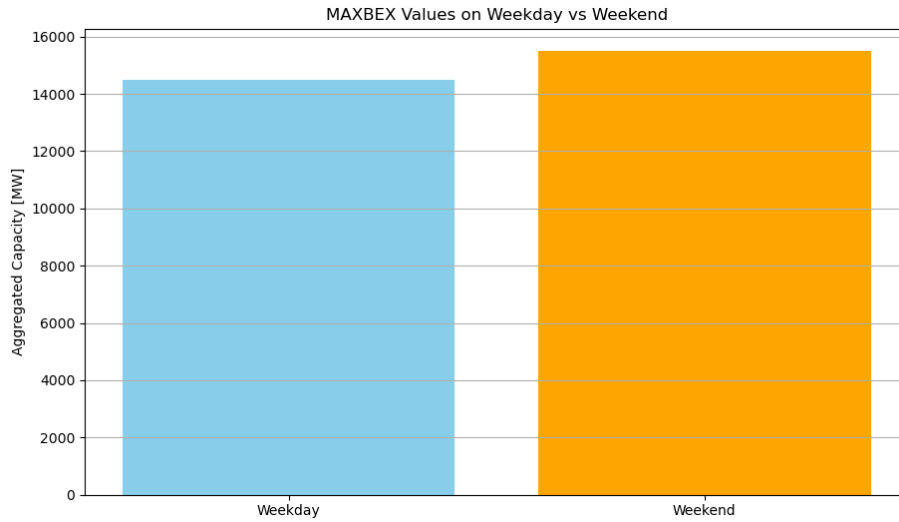


Figure 2.2: Aggregated MAXBEX values for three specific borders on weekdays and weekends.¹⁰

¹⁰MAXBEX borders: BEL_FRA, GER_POL, GER_FRA

3. Wind congestion: Germany limits export capacities when national wind generation is high. A study from Michael Bucksteeg et al. additionally supported this hypothesis [9].

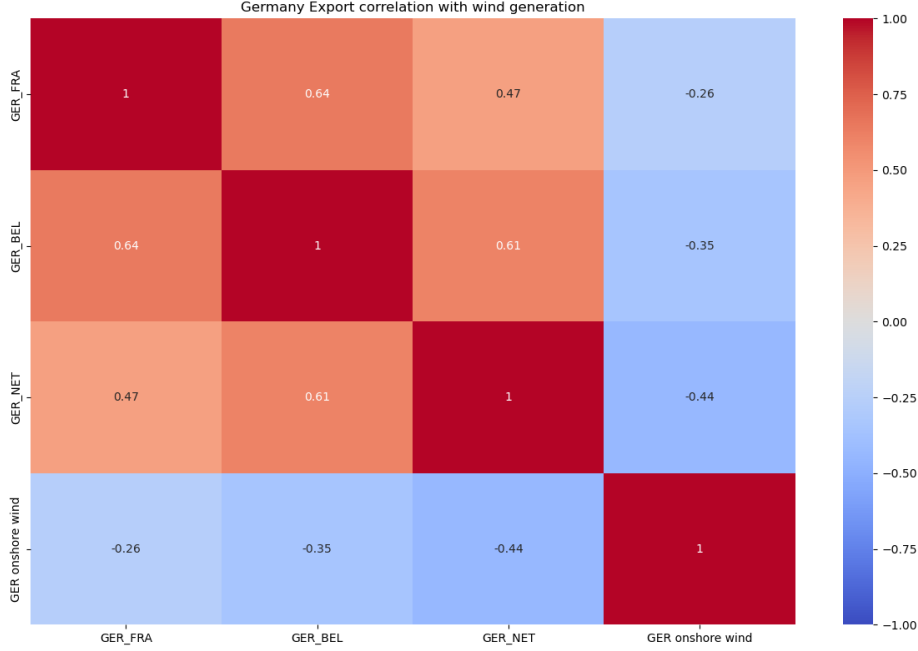


Figure 2.3: Correlation map of some of Germany's export borders and onshore wind generation in Germany.

There are a few hypotheses that were **falsified** during the process, two of them being:

1. Hydro congestion: Heavy rain results in high power generation concentrated in areas dominated by hydropower, which has a similar effect on the power grid to intense wind power production.
2. Overproduction in MAXBEX: The flow-based capacity allocation procedure should better account for expected inflexible power generation (hydro, wind, solar) patterns and allow for higher exports in countries with a large share of renewable energy production.

2.1.2 Feature Engineering

After validating the data, a feature map was designed. Following the rule of joint entropy in information theory "*information never hurts*", the raw data is retained in the dataset, where aggregated columns are added. Two datasets are created per border type (NTC and FBMC): A **baseline (BL)** and a **feature-extended (FX) version**, which are later used for training. The baseline dataset contains all raw data (in gray) and feature

columns (in yellow). The main difference between the two versions is the incorporation of electricity prices and border capacities from the past day and the past week.¹¹

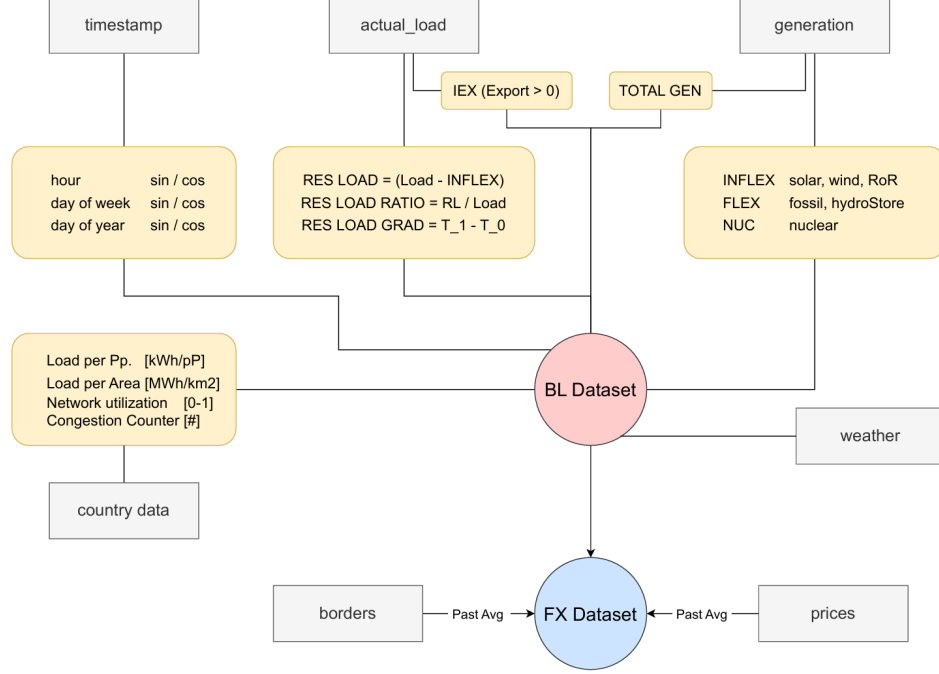


Figure 2.4: Feature engineering map with two resulting main datasets

The reason behind creating these two datasets is that the BL dataset is based solely on data that can be fed into the LPP model (Figure 1.1) to make a price forecast. In contrast, the FX dataset includes electricity prices as an output of the LPP model, and past cross-border capacities imply that an iterative approach is necessary to make future predictions, since these are originally the target variables of this project. This setup is intended to test whether it is feasible to predict interconnector capacities using publicly available data that is expected to have predictive power, or whether there is a deeper, more intrinsic connection between input data and cross-border capacities, as mentioned in the work of Abdel-Khalek et al. [2].

2.1.3 Column Filtering and PCA

To start with, a random forest analysis was performed on the dataset. After carefully evaluating and discussing the columns with high predictive power, it became evident that some features - despite scoring highly - should not logically influence predictions for certain borders. This led to a refinement of the dataset for each border using the

¹¹Past average electricity prices and border capacities were taken for the time window up to the present day, e.g. daily past averages for any time on May 21st were taken from May 20th 00:00 a.m up to May 20th 23:00 p.m.

neighboring country technique. If a border from A to B is picked, every country that is adjacent to A or adjacent to B is included in the selection. The picture below illustrates this filtering approach with 2 different examples. This decision was made to better accommodate the structural and statistical differences between the borders and deliberately neglect misleading input feature columns.

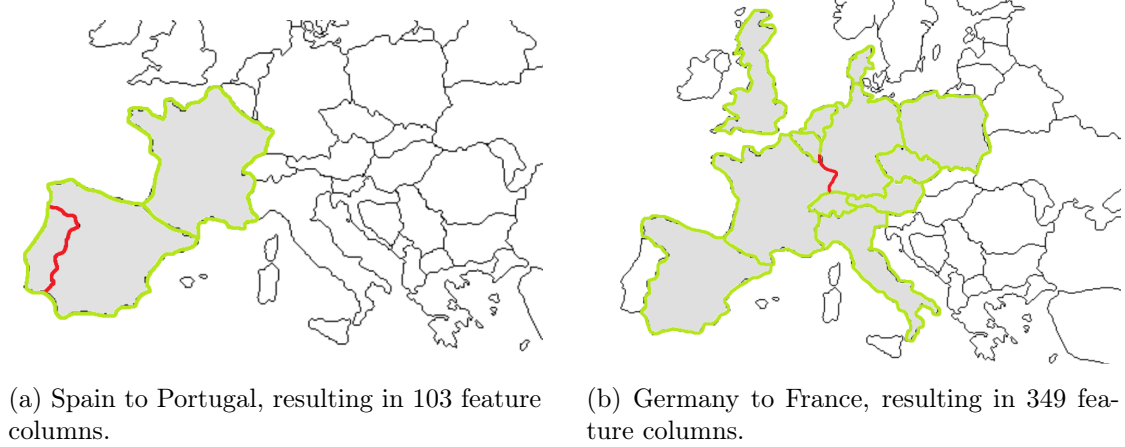


Figure 2.5: Illustration of neighboring country filtering for two different borders.

Secondly, after the columns were filtered, a principal component analysis (PCA) was performed, reducing the dimensionality for **each border-specific dataset** down to 64 dimensions, while always maintaining an explained variance of $> 93\%$ after PCA¹². Reducing the dimensionality of the input was mainly done due to computational resources.

2.1.4 Normalization

Target variables

Since the war in Ukraine, the demand for electricity does not increase as much as in the past years, as a consequence of declining industry production. Nonetheless, interconnector capacities are steadily growing, which is explained by an increasing renewable power generation as well as the target of reducing price differences across bidding zones. This creates a challenge for time-series modeling: If a major transmission upgrade occurs just after the training window ends, the model faces values in the validation and testing process that lie outside the distribution it was trained on. For that reason, a "*max-so-far*" normalization was applied to the target variable. This scales each value by the maximum seen up to that point in time, bounding the data between 0 and 1. In the figure below, an example with the cross-border capacity from Austria to Italy is depicted, which shows why this normalization is beneficial.

¹²Typically the number of components to retain a total variance of around 95% is selected, <https://vtantravahi.medium.com/principal-component-analysis-pca-37dc2c22cdf0>

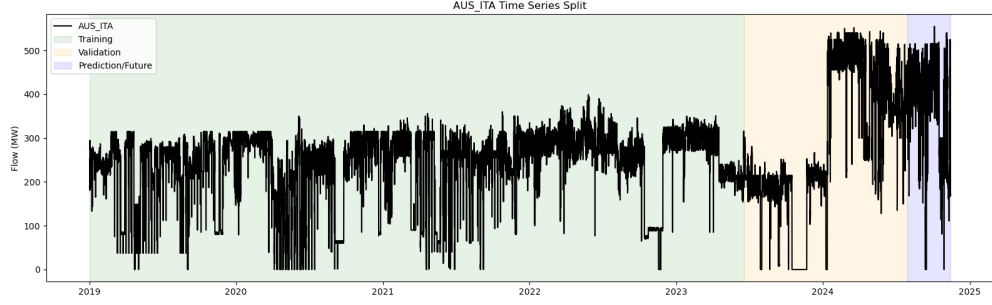


Figure 2.6: Border from Austria to Italy before "max-so-far" normalization.

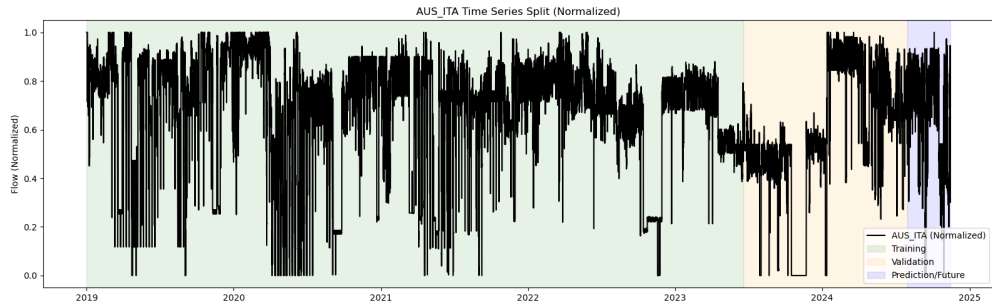


Figure 2.7: Border from Austria to Italy after "max-so-far" normalization.

This normalization technique is helpful for increasing values, but if the capacities decrease over time, there is no benefit. Secondly, the exact value may still be difficult to predict since there is no guarantee that the value has occurred in the training set. As visible in the two figures above, every target border was split into a training set, a validation set, and a testing set. Details on the split ratios are found in Table 3.1

Feature columns

Before reducing the input to 64 dimensions with PCA, different built-in scaler mechanisms have been tested from the popular scikit-learn library in Python. Due to the "max-so-far" approach chosen for the target variables, sklearn's MinMaxScaler was used to scale the feature columns, since it bounds the data to the same range from 0 to 1 by default.

2.2 Modeling

Different model architectures are used to analyze differences and develop an understanding of which aspects in certain architectures enhance the prediction quality. All models were trained using a **single target approach**, where a specific dataset is created for each border using the neighboring country technique shown in Figure 2.5.

Table 2.2: Overview of models

Model Name	Functionality
Base model	Simple, used mainly for benchmarking
Net	Multilayer perceptron, ReLU activation, dropout
LSTM	Long short-term memory, designed for time series modeling
Hybrid	Border specific (classification, regression)

The choice of the four listed models was made to have a simple referencing model, the base model, which set a minimal baseline to benchmark on. Secondly, the neural net is the natural expansion of the base model, which served the purpose of observing how naive fully connected layers perform in the task of predicting complex real-world cross-border capacities. Next, the LSTM, which is a specific type of recurrent neural network (RNN) specifically designed to handle sequential data and capture long-range dependencies, therefore interesting to compare against the other models. Lastly, the hybrid model differentiates between regression and classification depending on the border. This approach was chosen by taking a closer look at border capacities over time; additionally, Anna Fragkioudaki et al. [10] used a similar approach to predict day-ahead price spikes using regression and classification trees.

2.2.1 Regression Models

The most simple approach that one can think of is to directly map the input to the output layer, which is exactly how the base model architecture is built. The net is the extension of the base model by adding fully connected hidden layers, activation functions (ReLU), and a dropout chance for the nodes on the last couple of layers. The figure below illustrates these two models.

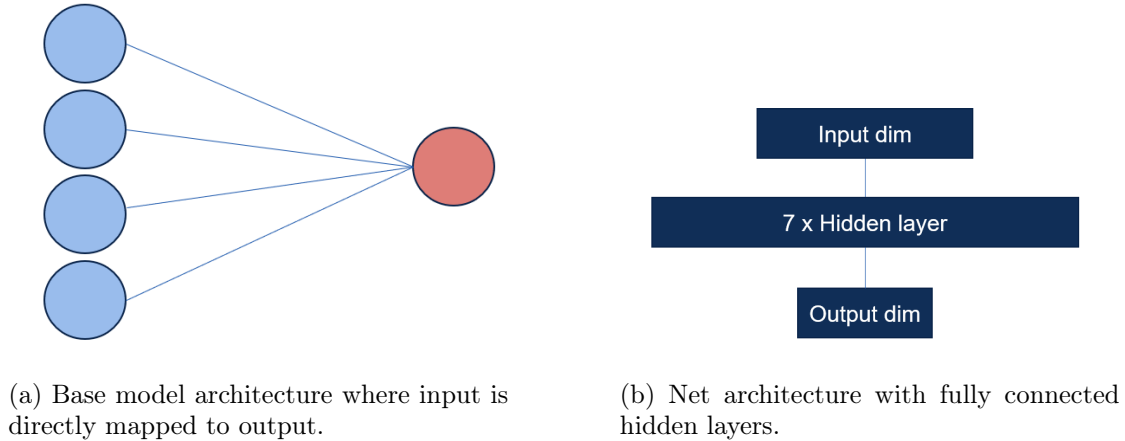
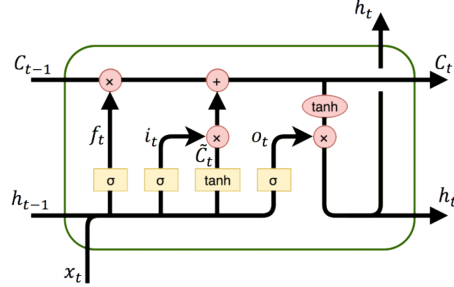


Figure 2.8: Comparison of base model and net architecture

With its ability to save temporary cell states and memorize a time sequence, the LSTM model is specifically designed to model time series. The next figure shows the standard LSTM architecture used in this project.



Variable Descriptions:

- $x(t)$: Input sequence at time t
- $h(t-1)$: Hidden state at $t-1$
- $C(t-1)$: Cell state at $t-1$
- $C(t)$: Updated cell state
- $h(t)$: Updated hidden state

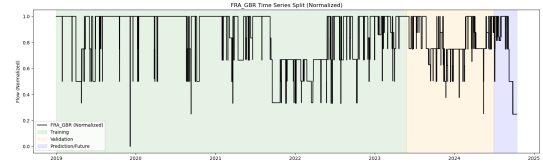
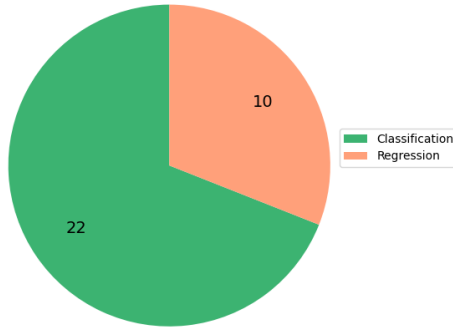
Figure 2.9: Schematic diagram of LSTM model architecture.¹³

2.2.2 Hybrid Model

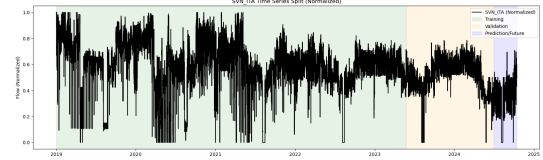
Regression and Classification Borders

Since the FBMC-Zone capacities are the result of an overall system optimization, while the NTC values are determined bilaterally between the corresponding TSOs of the countries, this leads to a less diversified and more discrete capacity distribution in the NTC domain. This brought about the idea of using the hybrid model, distinguishing if the border falls into a regression or a classification approach. The figure below underscores the difficulty of dividing the borders into these two groups and that a simple FBMC $\leftrightarrow$ NTC distinction is not sufficient.

Border division in NTC domain



(a) France to Great Britain (NTC)



(b) Slovenia to Italy (NTC)

Figure 2.10: NTC border differentiation for hybrid model.

¹³<https://apmonitor.com/pds/index.php/Main/LongShortTermMemory>

Separating all borders into classification and regression with statistical measures was unsuccessful¹⁴. For that reason, the borders were manually separated. A detailed breakdown is listed in the appendix in Table A.1.

As shown in Figure 2.10, most of the NTC borders are solved by classification. For these borders, classes were created by looping through the entire time series, and every unique value corresponds to a class which is in the end chosen through a **softmax** normalization by the cross-entropy loss. Note that all borders in the FBMC domain are still solved with the regression part of the hybrid model.

Hybrid Model Architecture

After the input goes through a projection layer, it is processed by three residual blocks. Secondly, the border task type is checked, and finally the latent space is passed through three fully connected layers. The following picture illustrates this architecture.

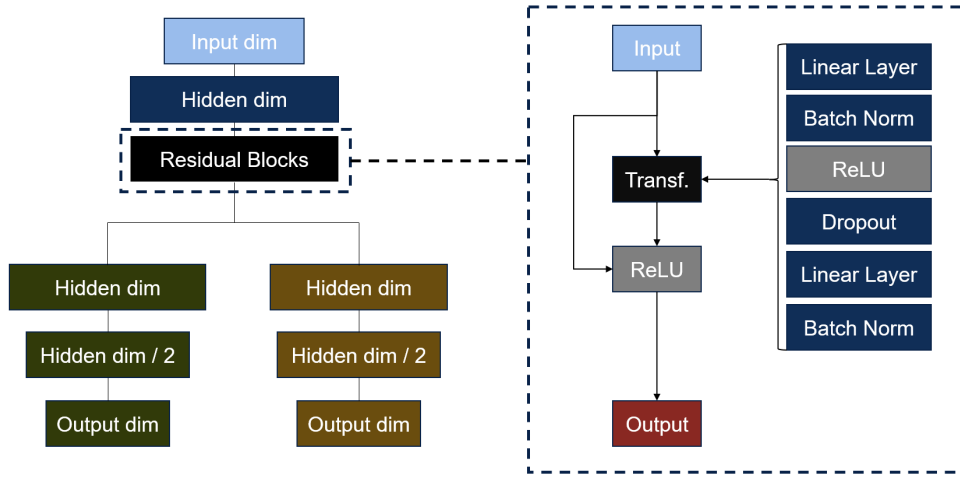


Figure 2.11: Hybrid model architecture with detailed view on residual block structure.

The main difference to the neural net shown in Figure 2.8b are the highlighted **residual blocks** used in the core of the hybrid model, which demonstrated improved forecast precision for time series modeling in the past [8], [11]. These residual connections allow the model *"to do nothing"*, even though the input variables may have changed, because the input passes through a series of fully connected layers with an activation function, but is added again in raw form before the last activation function.

¹⁴Metrics such as mean, standard deviation, gaps between consecutive hours, or average step size were analyzed

Chapter 3

Results

3.1 Training Conditions

To keep training runs on different models and different datasets consistent throughout, the following results are all carried out with a fixed seed¹⁵ and hyperparameters listed in the table below.

Table 3.1: Training hyperparameters for all models

Parameter	Value	Hybrid (Grid Search) ¹⁶
Epochs	40	-
Batch size	128	32
Learning Rate	3×10^{-4}	3×10^{-3}
Weight Decay	1×10^{-4}	-
Dropout	40%	-
Hidden dimension	128	256
Criterion ¹⁷	MSE / Cross Entropy	-
Train/Test Split	95% of full dataset	-
Validation Split	20% of training dataset	-
Feature Scaling	MinMaxScaler before PCA	-
Target Scaling	"max-so-far"	-
Training Type	Single target	-
NTC	32 Borders	-
FBMC	19 Borders	-

¹⁵Six different seeds converged to comparable values after a similar amount of iterations.

¹⁶Due to computational reasons a grid search has been performed after all training runs for the best performing model only.

¹⁷Mean Squared Error (MSE) loss was chosen for the regression to prioritize the overall trend with a smooth gradient. Cross Entropy (CE) loss was used for classification to model the probability distribution over discrete capacity values.

3.2 FBMC Results

Training losses

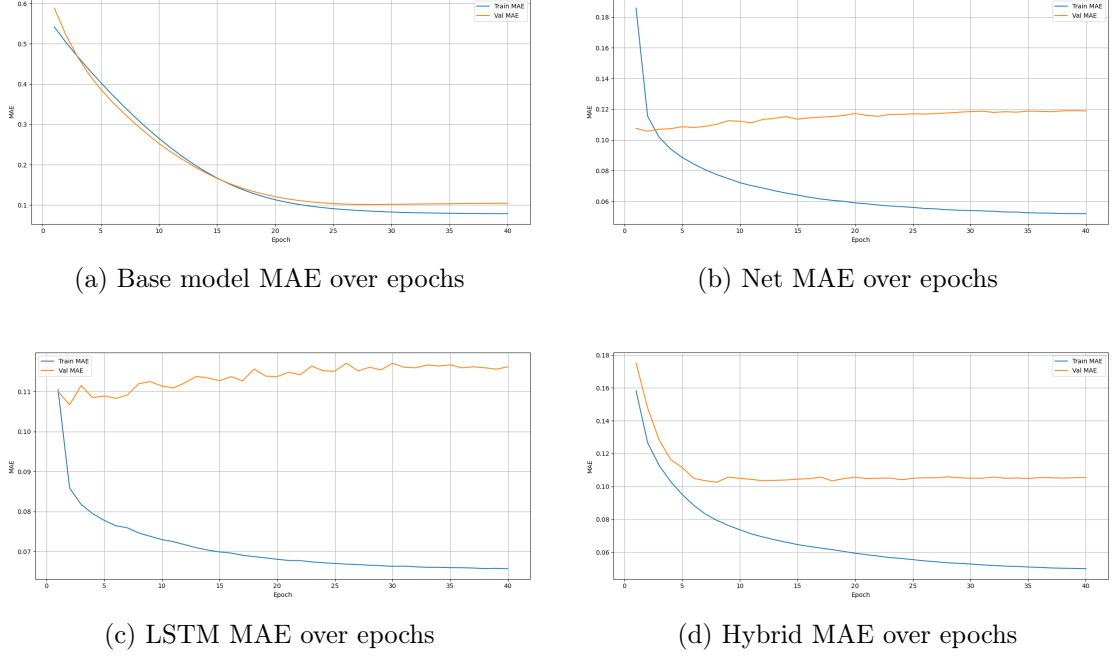
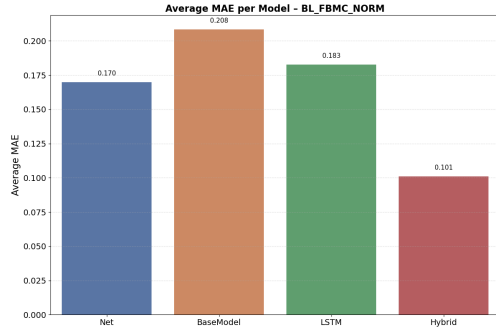


Figure 3.1: Average training MAE (blue) and validation MAE (orange) over time for different models on FBMC borders (full sized in the Appendix).

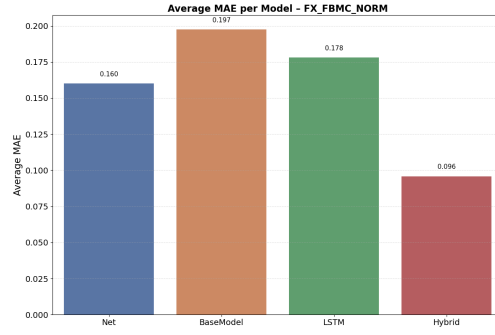
All models can be trained from the input features. Although, taking a closer look at the neural net and LSTM they fail to generalize learned parameters on unseen data, as the validation curve even increases slightly over time. However, all models end up with a similar MAE score after 40 epochs. Due to the deeper complexity of the neural net and LSTM, they immediately capture simple relationships other than the base model. With the usage of residual connections, the hybrid model takes some more iterations to capture these dependencies in the data. Note that there is a drastic change in the behavior of the cross-border capacities, starting in the second half of the year 2024, which was chosen for validation and testing, and potential reasons for such scores are discussed in more detail in Section 4.1.1.

Testing metrics

While FBMC borders exhibit continuous behavior, especially the LSTM model, which develops a memory, it was expected to catch temporal patterns. In the following comparison, MAE is averaged over all borders for each dataset.



(a) Average MAE in testing on BL dataset



(b) Average MAE in testing on FX dataset

Figure 3.2: Average MAE in testing for each model on BL and FX dataset (full sized in the Appendix).

It is clearly visible that the hybrid model scores best out of all models. Secondly, the predictions on the FX dataset are slightly better on average for every model with a lower MAE of around $\approx 5\%$. The residual blocks are certainly the cause of drastic improvement and better ability to generalize on unseen data. Next, a more detailed view per border on the R^2 -scores is provided to illustrate which borders are easier or harder to predict.

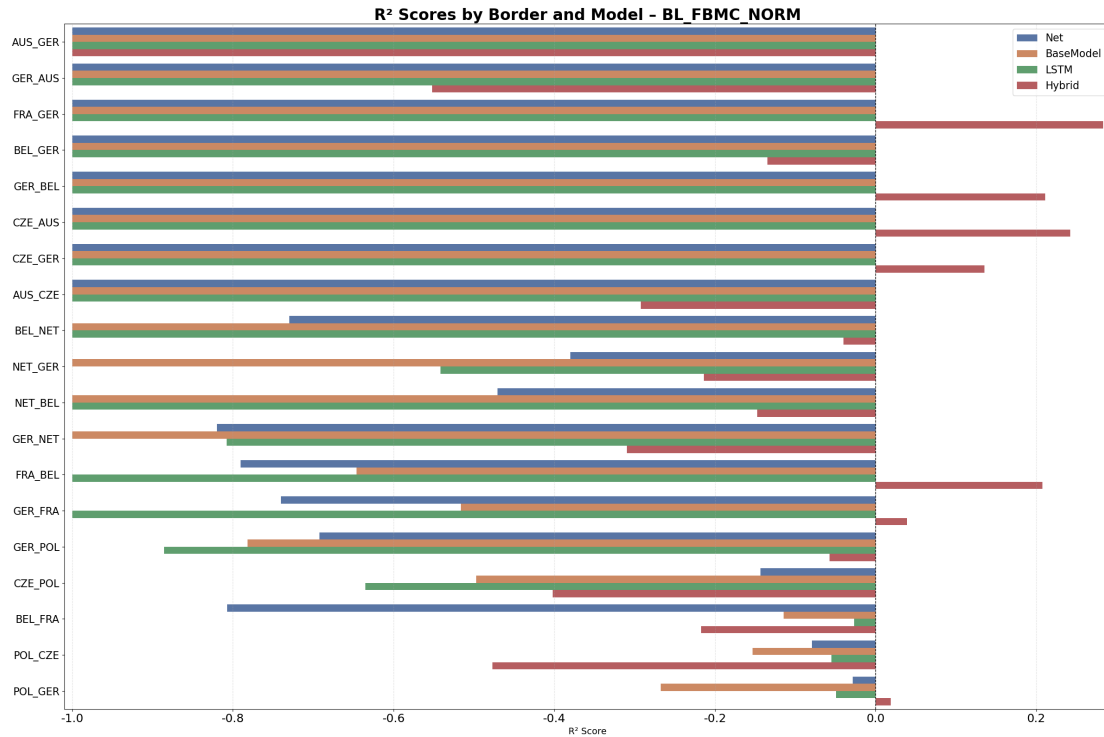


Figure 3.3: Per border R^2 -scores for each model on BL dataset.

The hybrid model is the only one to score positive R^2 -Scores on some of the FBMC borders. The others fail to generalize learned parameters on unseen data. Note that the R^2 -Score is capped at -1 in Figure 3.3, so there are several borders where the models do not predict reasonable values. Taking a closer look at one such example, namely the border from Germany to Austria, it becomes clear why some of the predictions score so badly.

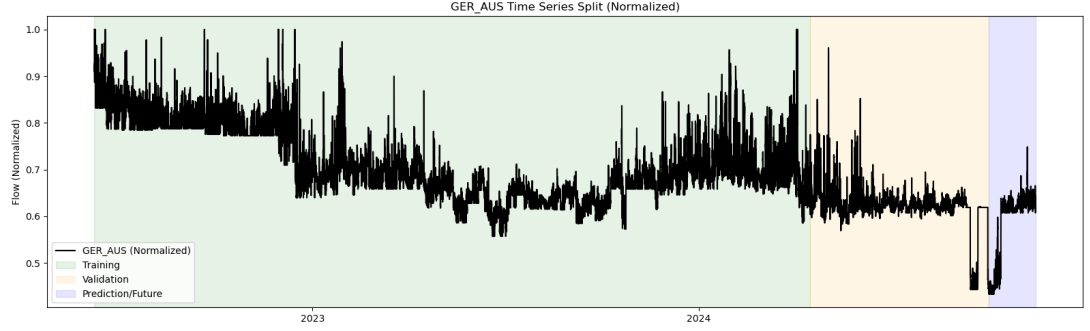


Figure 3.4: Germany to Austria border with training, validation, and test split.

During training (green), the model tries to predict values that do not exist during validation (orange) or testing (blue). The "max-so-far" normalization shown in Figure 2.7 does not help here, since the values decrease over time.

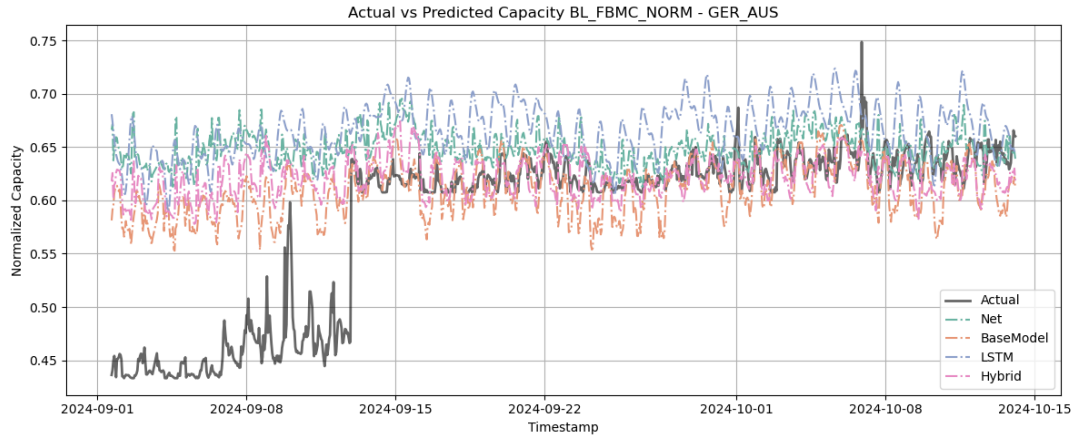


Figure 3.5: Predictions of every model on the border from Germany to Austria.

Despite failing to predict the first third of the ground truth, seasonal and daily patterns, and the overall mean of the predictions is captured best by the hybrid model. Possibilities to depict such events are discussed in more detail in Section 4.1.2.

3.3 NTC Results

Training losses

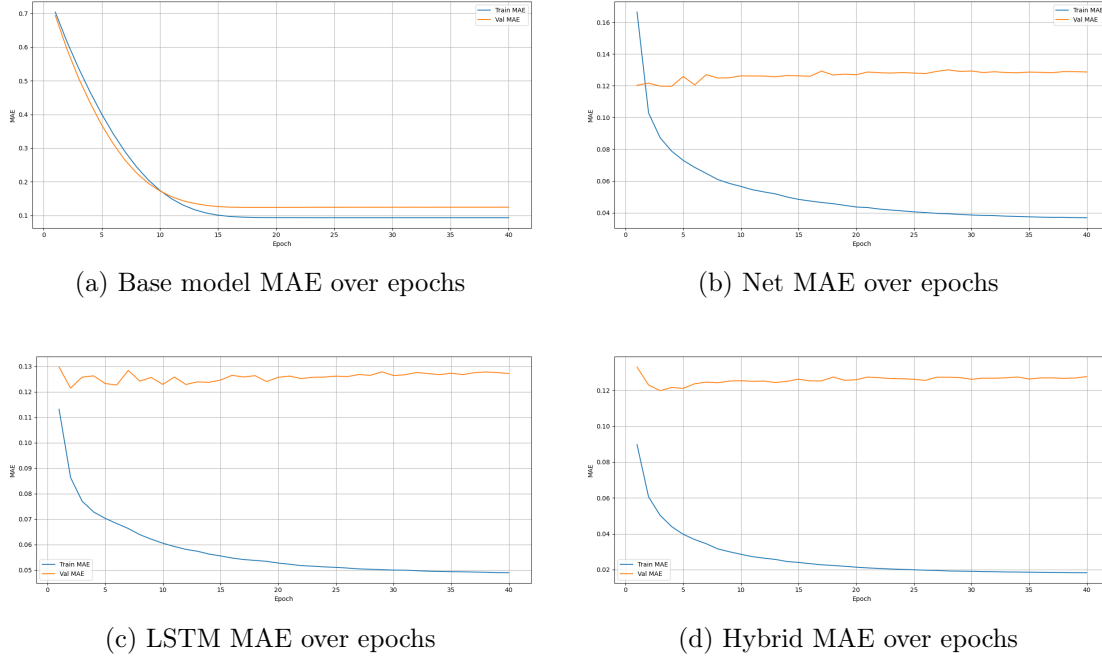
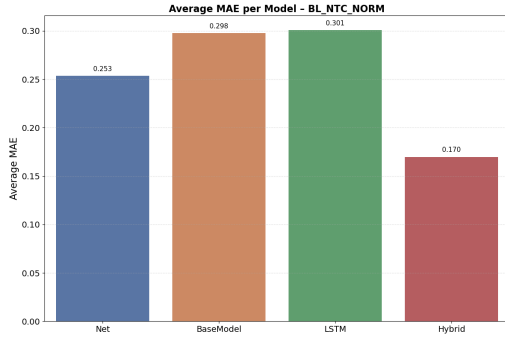


Figure 3.6: Average training MAE (blue) and validation MAE (orange) over time for different models on NTC borders (full sized in the Appendix).

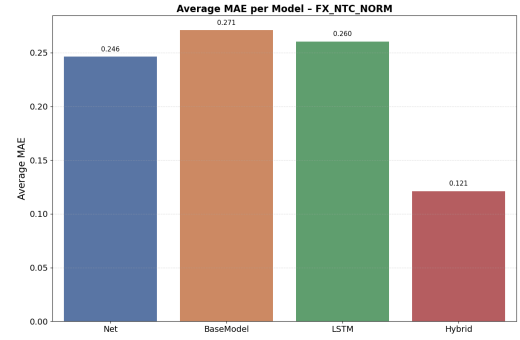
Similarly to the FBMC borders, the models are able to train (blue) from the input data, and capture patterns very fast due to the deeper architecture compared to the base model. The hybrid shows a worse generalization than in the FBMC domain, when looking at the Validation MAE (orange) over all borders. This is likely due to the characteristic solving process of classification, where picking a wrong class can induce a higher loss than swerving away from the ground truth as in regression. It is important to mention that with classification, the MAE has higher variance across borders, especially when compared to a regression model. Again, the problematic behavior of some borders from late 2024 onward poses a problem for validation scores.

Testing metrics

For the NTC borders, expectations were towards the hybrid model due to the classification approach. Although some of the borders are still solved with regression, the hybrid model outperformed the others in regressional borders as well. In the following comparison, MAE is averaged over all borders for each dataset.



(a) Average MAE in testing on BL dataset



(b) Average MAE in Testing on FX dataset

Figure 3.7: Average MAE in testing for each model on BL and FX dataset.

Analogously to the FBMC borders, the hybrid model scores the lowest MAE when averaged over all borders, lowering the MAE on the feature extended dataset (FX) scores lower MAEs for every model, which confirms the predictive power of past electricity prices and border capacities. To better understand the performance of each model, a detailed view per border of the MAE scores is provided below.

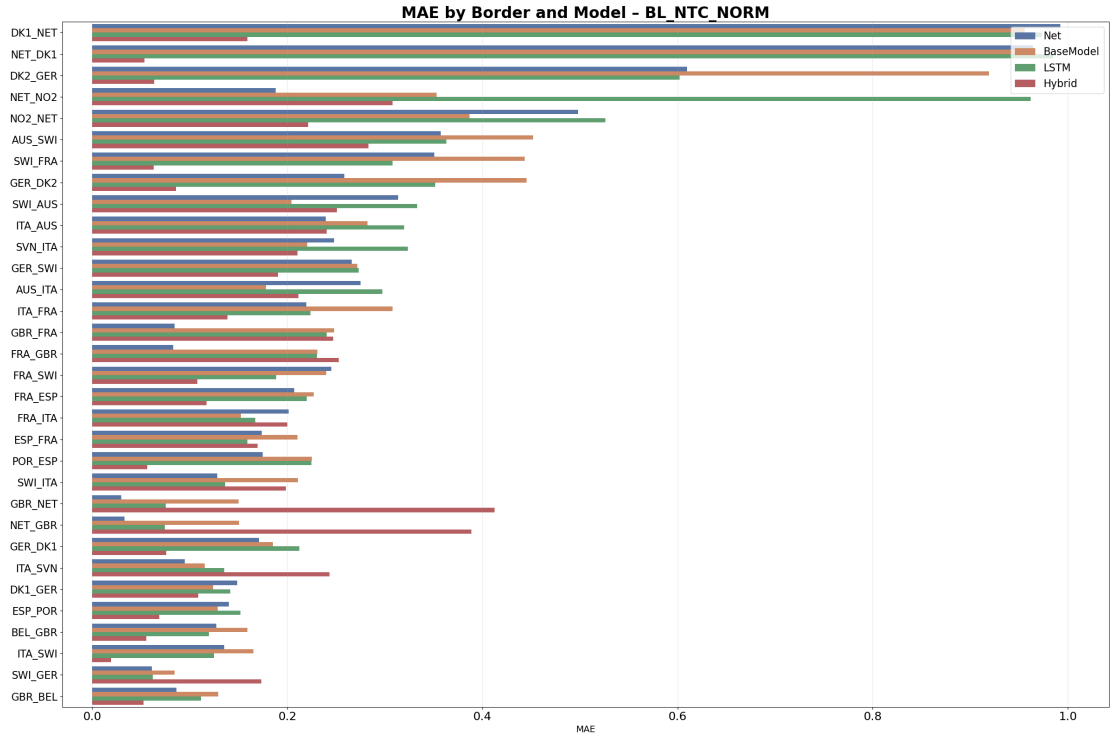


Figure 3.8: Per border MAE scores for each model on BL dataset.

Although the hybrid model performs best on average over all borders, there are some borders which are predicted better by the *regression-only* models. Taking a closer look on which borders are predicted worse by the hybrid model, it becomes clear that these are all borders the model uses classification on, where the loss from a certain timestep can be much higher since the prediction jumps between values other than fitting a smooth curve on it. Nevertheless, to emphasize the strength of the classification, a closer look is taken at the best prediction of the hybrid model with the border from Italy to Switzerland in the figure below.

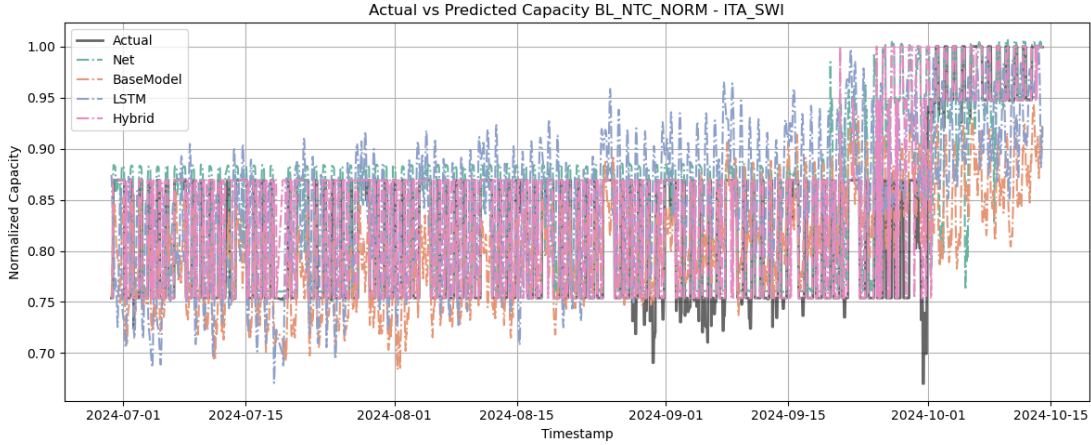


Figure 3.9: Predictions of every model on the border from Italy to Switzerland.

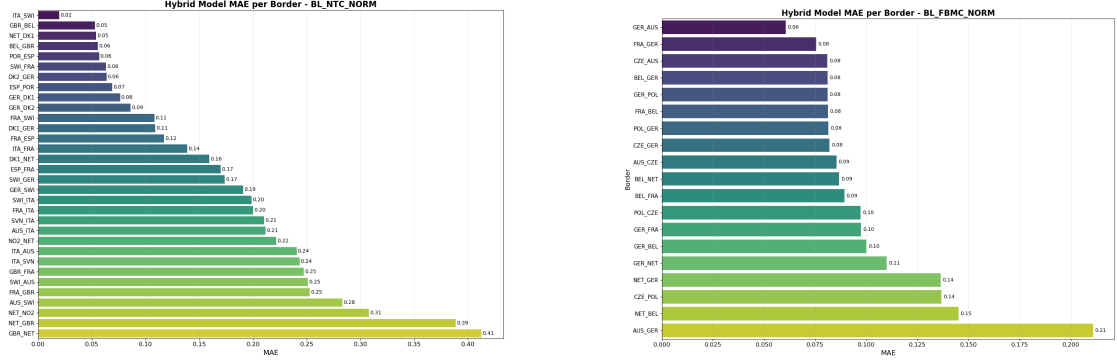
Observing how the three regression models slightly diverge from the ground truth, the classification approach of the hybrid model delivers an MAE of around 0.02 with an accuracy of 71%. Hence, some borders definitely benefit from this hybrid approach, while on other borders, which still exhibit *"stepwise"* behavior and are categorized as classification borders, would be better solved through regression. However, it is difficult to determine the border division in advance, by only looking at the border capacity values on a timescale. A more detailed interpretation and different approaches to refine this division are found in Section 4.1.3

3.4 Detailed Hybrid Model Analysis

Because the hybrid model achieved the best MAE scores among the others, as shown earlier in Figures 3.2 and 3.7, certainly due to the implementation of residual layers, a detailed analysis of the model follows for a better understanding of its performance and capabilities.

Testing Metrics

In comparison below, MAE is shown on all borders for the BL datasets.



(a) Hybrid model MAE on NTC borders

(b) Hybrid model MAE on FBMC borders

Figure 3.10: Overview of MAE on all borders for hybrid model (full sized in the Appendix).

In general, it can be observed that FBMC borders fluctuate less in MAE than NTC borders, which can be explained by the model's classification solver, where a wrong class tends to lead to higher errors than fitting a smooth curve, since, for example, the borders (GBR_NET, NET_GBR, NET_NO2, AUS_SWI, DK2_GER) with the **five highest MAE on average are all classification borders** in the NTC domain.

3.4.1 Backtest

To validate the predictions of the hybrid model, a backtest has been performed, where the second half of 2023 was left out for testing. The resulting predictions were fed into the LPP-model together with BKW's standard cross-border profiles and historical NTC values for comparison. A schematic overview of the process is depicted below.

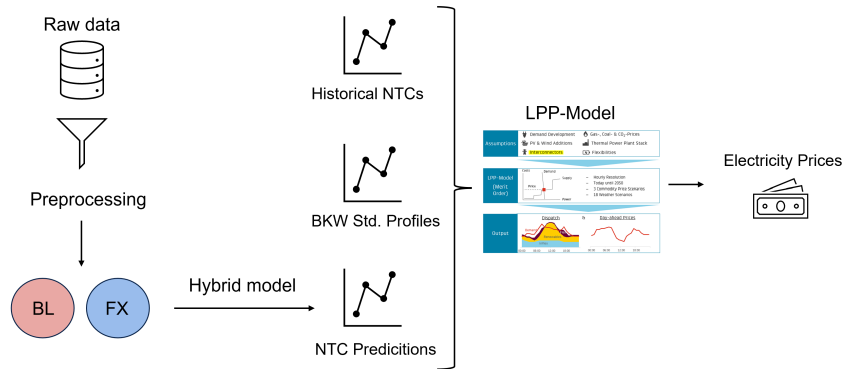


Figure 3.11: Schematic flowchart of backtest process performed by BKW.

Because the predicted MAXBEX values in the flow-based domain represent the maximum possible cross-border capacity, given that all other net positions are 0, they have to be scaled down to match the actual capacity. Since BKW does not have a qualitatively good enough model for scaling down these MAXBEX values, only NTC predictions are changed in the backtest, whereas FBMC values are kept constant throughout, to eliminate possible scaling errors. The following figures show the hybrid model performance on the BL and FX dataset compared to BKW's current NTC prediction strategy, and actual historical NTC values by calculating the electricity price across all bidding zones with the LPP-Model.

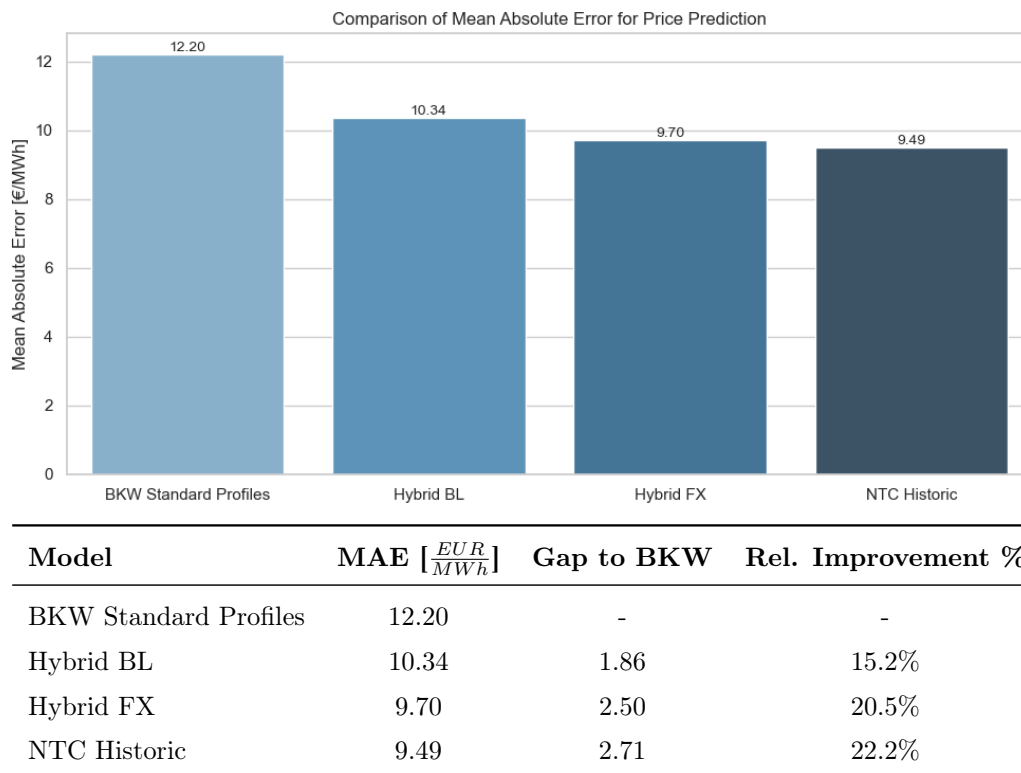


Figure 3.12: Comparison of MAE across models, showing improvement over BKW's standard profiles.

There are two main reasons for the high MAE in the electricity prices even after using historical NTC values. First of all, the simple assumptions for the FBMC profiles used throughout all runs, where some borders utilize wind correlations to predict capacities, but others are simple flat line profiles. Additionally, electricity prices in the Nordic countries (Sweden, Norway, Finland) which are hydro-dominated and difficult to predict, are split up into multiple bidding zones and pose a significant contribution to the MAE.

However, feeding historical NTC values into the LPP-Model results in the lowest MAE for price predictions, verifying the validity of the predicted results from the hybrid model. With the BL dataset, electricity price predictions already gain an improvement of 15.5%, which is significant when compared to the 22.2% improvement using historical NTC values. Furthermore, the FX dataset delivers predictions very close to the historical values, which was expected since past averaged historical values are used.

Chapter 4

Discussion

4.1 Problems

4.1.1 Real-World Events

Taking a closer look at the cross-border capacities in Europe during the time window from January 2019 to December 2024, some borders exhibit abnormal behavior from 2024 or even earlier. Real-world events, such as energy crises, political decisions, or environmental hazards, have a significant impact on cross-border capacities. Given their unpredictability, such phenomena pose challenging circumstances for AI-based models during training and testing.

In this case, the prediction time window was set in the last couple of months of the year 2024, which initially appeared reasonable, as using the most recent values is a common practice in time series forecasting, circumventing the danger of data leakage. However, if the target variable changes drastically before or during the testing window, it is difficult to determine a meaningful inference from the result. A clear example of where this is the case is shown in Figure 2.6 where the cross-border capacity from Austria to Italy suddenly increases in 2024, making it difficult for a trained model to predict this radical change.

Designing the model resilient to changes led to the implementation of residual connections, which improved the ability of the model to generalize on unseen data. Therefore, building on top of **pre-trained residual networks**, such as LassoNet or NBeatsX, which were used to predict electricity prices [8], promises even greater robustness.

4.1.2 Feature Columns

Another problem addressing the stability of the model is the feature selection. In the beginning, a random forest analysis was performed to analyze the predictive power of certain columns, and because it was obvious that certain columns should not have a high predictive power for the cross-border capacities, the filtering approach for neighboring countries was implemented, as illustrated in Figure 2.5. However, this eliminates columns from the scope of the training data that would otherwise have had an impact. Finally,

since real-world events influence the target variable, feature columns such as network utilization and network congestion were implemented, which also showed great predictive power when running a random forest analysis. Therefore, further investigation of possible feature columns is needed, to depict, e.g., power plant outages or political changes, similarly reported by Abdel-Khalek et al.[3], to deepen the understanding of the model for cross-border capacities.

4.1.3 Border Division

Finally, as important as picking a tailored selection of features is picking the correct division of borders when using a hybrid solver. First of all, the classification approach is definitely beneficial, especially for borders in the NTC domain, such as the border from Italy to Switzerland shown in Figure 3.9, where the cross-border capacity was predicted with a MAE of 0.02, whereas on other borders, such as Great Britain to the Netherlands (NTC border, solved with classification for the hybrid model), the base model has a lower MAE than the hybrid model. This poses the question of whether the borders are separated correctly or if another division would yield better results. In this case, an **iterative approach** would be possible where each border is tested in a classification solver and a regression solver to compare the results.

4.2 Conclusion

4.2.1 Baseline and Feature Extended Dataset

Introducing past averages of the targeted cross-border capacities and electricity prices in the FX dataset was expected to yield more accurate predictions, which proved to be the case when compared to the BL dataset. Moreover, these past averages impacted the predictions in the NTC domain significantly, whereas for FBMC borders the improvement seems negligible. Nevertheless, looking at the backtest comparison in Figure 3.12, the improvement of 15.5% with the BL dataset and 20.5% with the FX dataset to BKW's current standard profiles is significant considering the theoretical limit of 22.2% when using historical NTC values, emphasizing the predictive power of publicly available data. Generally, the inclusion of past average electricity prices and border capacities improves the predicted values, but the **overall performance using only data compatible with the LPP-model appears sufficient for practical application.**

4.2.2 FBMC and NTC Borders

As mentioned, including past averages of target values for FBMC borders does not improve predictions significantly. However, predictions on FBMC borders yielded lower MAE values of around 35% with the BL dataset, and 30% with the FX dataset respectively, compared to the NTC borders when averaged over all four models. It becomes clear that the NTC domain remains more difficult to model, largely due to the lack of a standardized approach among TSOs for capacity determination, a challenge further

underscored by SwissGrid’s lack of transparency. Several complaints have been filed against Swissgrid AG due to the lack of transparency in setting cross-border capacities in Switzerland. As a consequence, a workshop has been arranged in 2024 to try to explain the different scenarios Swissgrid AG uses to determine interconnector capacities. However, more and more countries are switching to the FBMC-Zone, creating a setting where AI-based models can be trained with higher success. However, with only NTC borders being used in the backtest performed by BKW, it would be interesting to observe the performance of the model once a reasonable scaling approach for the MAXBEX values has been established. In general, **predictions on FBMC borders yield more precise values**, since the underlying mechanism to determine those values is more consistent. **Certain NTC borders continue to present significant modeling challenges** and may benefit from alternative model architectures or targeted feature enhancements.

4.3 Outlook

4.3.1 Modeling Improvements

Temporal Convolutional Network

Besides using pre-trained residual networks, another interesting approach is to use a temporal convolutional network, which enhances the ability to capture long-term patterns in time series while keeping the possibility of using residual connections. A preliminary approach to this idea has been tested, but due to the quickly increasing computational time, the low complexity of the implemented convolutional layer led to very similar results achieved by the hybrid model. The illustration below shows the 1D-convolutional layer used in the preliminary approach.

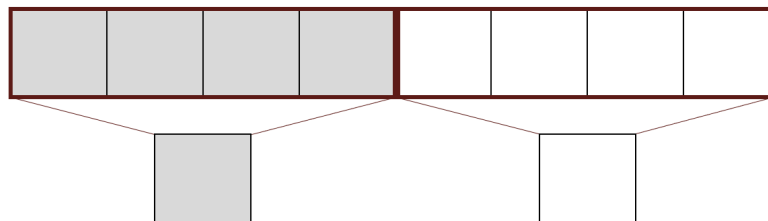


Figure 4.1: Schematic overview of temporal convolutional layer without dilation

Multiple convolutions can be stacked on top of each other, with each resulting block being a learned weighted sum, but using a dilation > 1 quickly increases the required sequence length of the input, leading to longer computation times.

Hierarchical Latent Space

Furthermore, the implementation of a hierarchical latent space, where the latent variable is extracted from different stages of the network and concatenated in the end, is similar to the idea of visualization mechanisms of node weights to understand which part of the input and, more importantly, after which transformation the model relies most on.

4.3.2 Data Expansion and FBMC Focus

Eventually, as multiple countries switched to the FBMC-Zone recently, and more transitions are expected to happen in the future, establishing a consistent set of target variables narrowed the time window used for training. Therefore, the FBMC cross-border capacities used to train for started in June 2022, while the NTC target values started in January 2019. It seems that a time series of around two years is sufficient to train a model; it certainly will help if the training is performed in cyclical time periods to observe if and how much adding more data to train on improves the resulting predictions. Another aspect is to focus more on FBMC borders and regressional models, leaving the hybrid approach as a temporary solution, since it is expected that more and more countries will switch to the FBMC-Zone.

Bibliography

- [1] Marten Ovaere et al. “The effect of flow-based market coupling on cross-border exchange volumes and price convergence in Central Western European electricity markets”. In: *Energy Economics* 118 (2023), p. 106519.
- [2] Goulven Salic and Yann Rebours. “Impact of German wind generation forecasts on net transfer capacities”. In: *2011 8th International Conference on the European Energy Market (EEM)*. 2011, pp. 635–640.
- [3] Hazem Abdel-Khalek et al. “Forecasting cross-border power transmission capacities in Central Western Europe using artificial neural networks”. In: *Energy Informatics* 2.1 (Sept. 2019), p. 12.
- [4] Yago Saez et al. “Integration in the European electricity market: A machine learning-based convergence analysis for the Central Western Europe region”. In: *Energy Policy* 132 (2019), pp. 549–566.
- [5] Bashir Bakhshideh Zad et al. “Supervised learning-assisted modeling of flow-based domains in European resource adequacy assessments”. In: *Applied Energy* 325 (2022), p. 119875.
- [6] Marie Girod et al. “Bid filtering for congestion management in European balancing markets à A reinforcement learning approach”. In: *Applied Energy* 361.C (2024).
- [7] Colin Webb and Hazem Karbouj. “Forecasting the Risk of Congestion”. In: *Project Annual Progress Report Document* (2019).
- [8] He Jiang et al. “Probabilistic electricity price forecasting by integrating interpretable model”. In: *Technological Forecasting and Social Change* 210 (2025), p. 123846.
- [9] Michael Bucksteeg, Katrin Trepper, and Christoph Weber. “Impacts of renewables generation and demand patterns on net transfer capacity: implications for effectiveness of market splitting in Germany”. In: *IET Generation, Transmission & Distribution* 9.12 (2015), pp. 1510–1518.
- [10] Anna Fragkioudaki, Adamantios Marinakis, and Rachid Cherkaoui. “Forecasting price spikes in European day-ahead electricity markets using decision trees”. In: *2015 12th International Conference on the European Energy Market (EEM)*. 2015, pp. 1–5.
- [11] Boris N. Oreshkin et al. *N-BEATS: Neural basis expansion analysis for interpretable time series forecasting*. 2020.

Appendix A

Appendix

A.1 Detailed Border Division

Table A.1: Breakdown of borders into regression and classification

Border	NTC	FBMC	Reg	Cls
AUS_CZE		X	X	
AUS_GER		X	X	
BEL_FRA		X	X	
BEL_GER		X	X	
BEL_NET		X	X	
CZE_AUS		X	X	
CZE_GER		X	X	
CZE_POL		X	X	
FRA_BEL		X	X	
FRA_GER		X	X	
GER_AUS		X	X	
GER_BEL		X	X	
GER_FRA		X	X	
GER_NET		X	X	
GER_POL		X	X	
NET_BEL		X	X	
NET_GER		X	X	
POL_CZE		X	X	
POL_GER		X	X	
AUS_ITA	X		X	
ESP_FRA	X		X	
ESP_POR	X		X	
FRA_ESP	X		X	

(continued on next page)

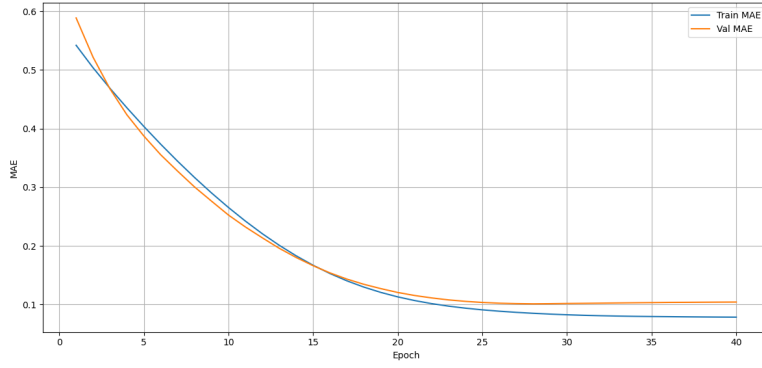
(continued from previous page)

Border	NTC	FBMC	Reg	Cls
DK1_GER	X		X	
FRA_ITA	X		X	
GER_DK1	X		X	
POR_ESP	X		X	
SVN_ITA	X		X	
SWI_ITA	X		X	
AUS_SWI	X			X
BEL_GBR	X			X
DK1_NET	X			X
DK2_GER	X			X
FRA_GBR	X			X
FRA_SWI	X			X
GBR_BEL	X			X
GBR_FRA	X			X
GBR_NET	X			X
GER_DK2	X			X
GER_SWI	X			X
ITA_AUS	X			X
ITA_FRA	X			X
ITA_SVN	X			X
ITA_SWI	X			X
NET_DK1	X			X
NET_GBR	X			X
NET_NO2	X			X
NO2_NET	X			X
SWI_AUS	X			X
SWI_FRA	X			X
SWI_GER	X			X

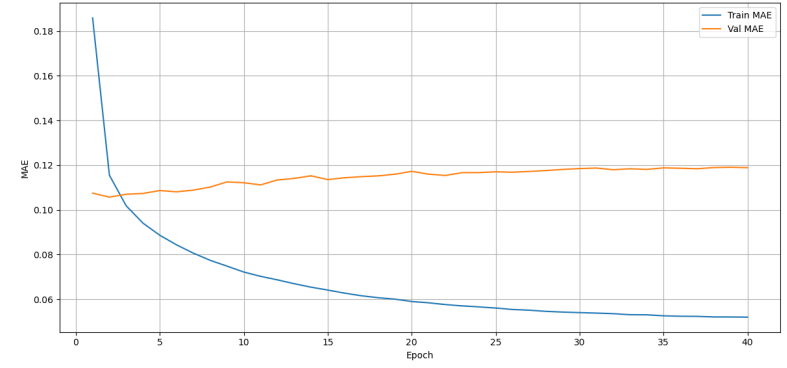
A.2 GitHub Repository

Code base build in Python and all figures can be found here:
<https://github.com/beandrej/XBorderCap-ML>

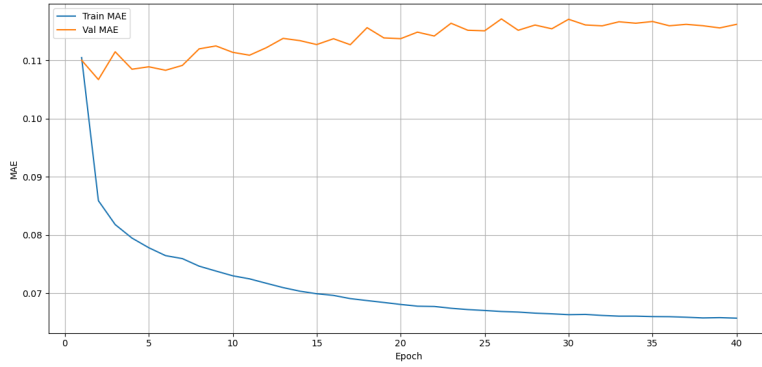
A.3 Figures



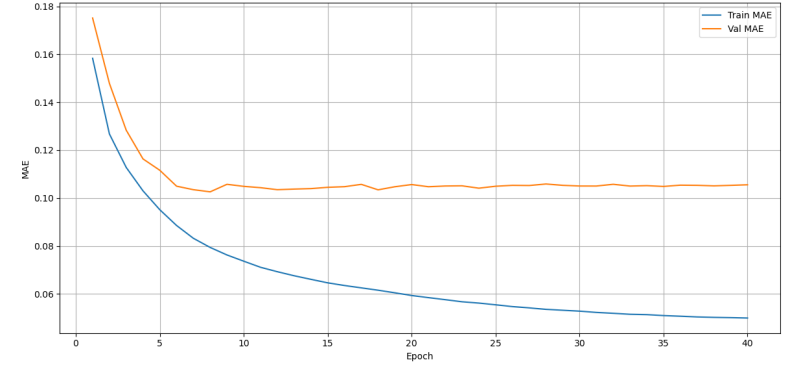
(a) Base model MAE over epochs



(b) Neural net MAE over epochs

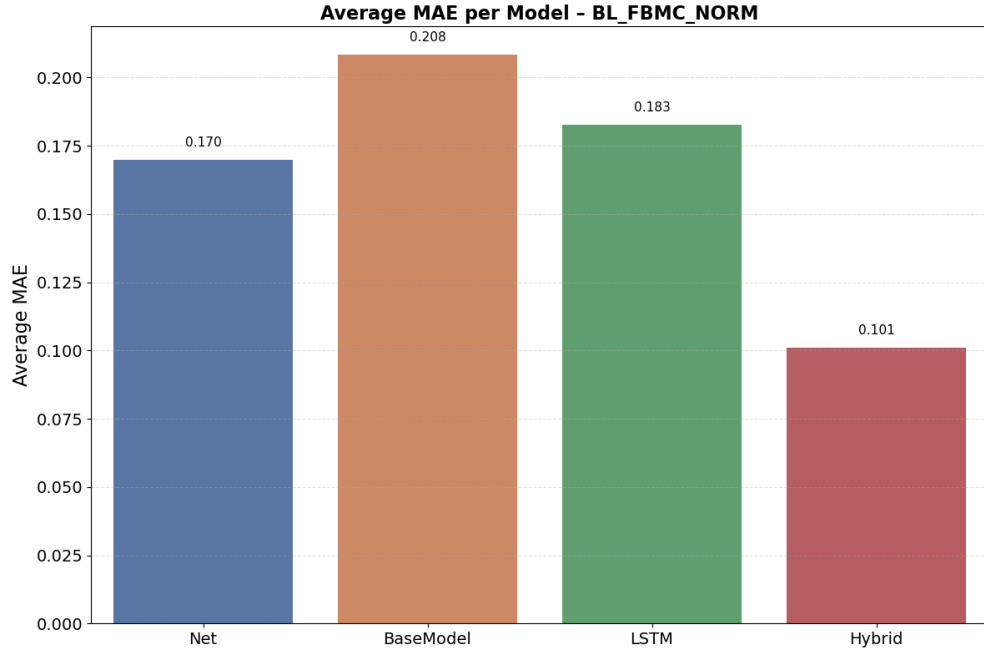


(c) LSTM MAE over epochs

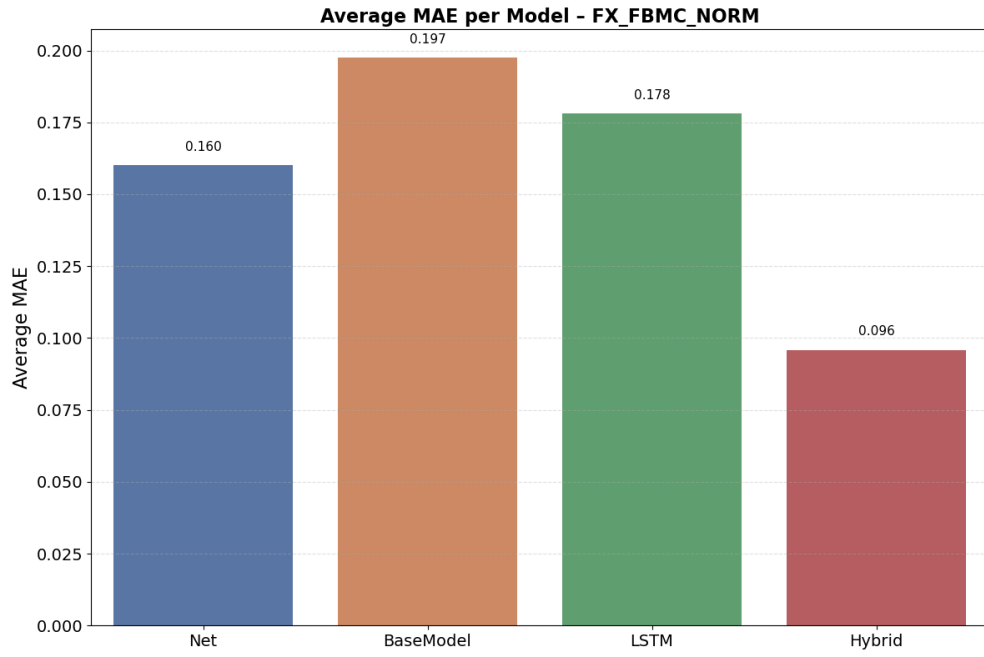


(d) Hybrid MAE over epochs

Figure A.1: Average training MAE (blue) and validation MAE (orange) over time for different models on FBMC borders.

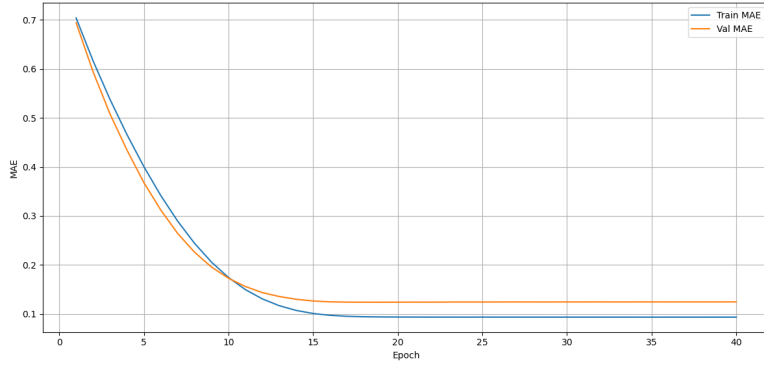


(a) Average MAE in testing on BL dataset

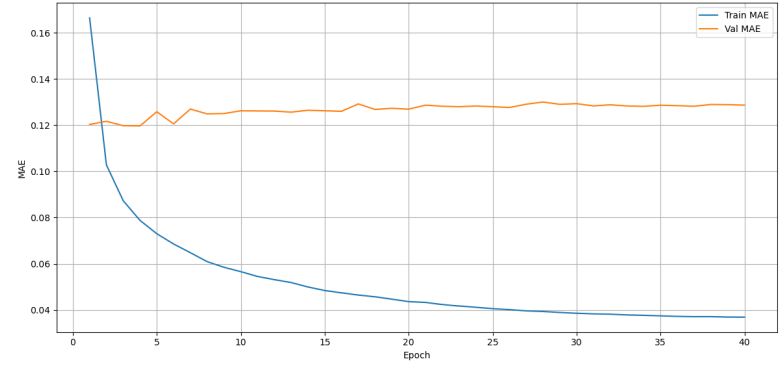


(b) Average MAE in testing on FX dataset

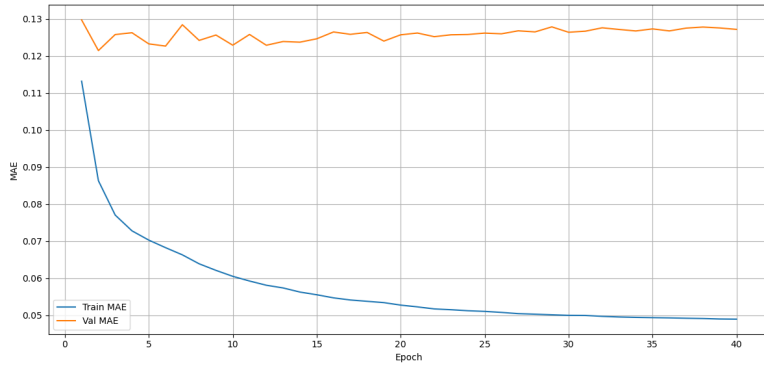
Figure A.2: Average MAE in testing for each model on BL and FX dataset.



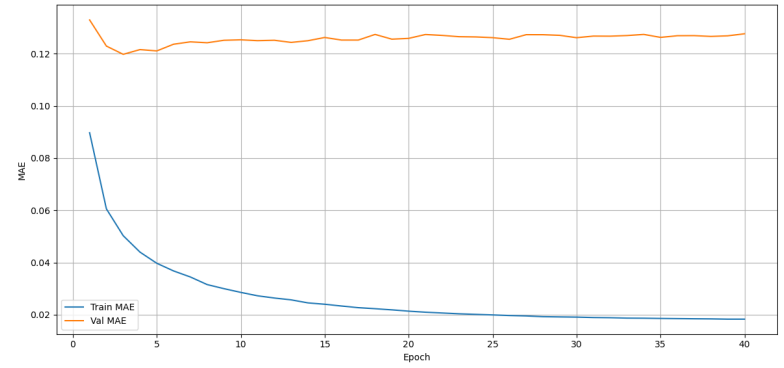
(a) Base model MAE over epochs



(b) Neural net MAE over epochs

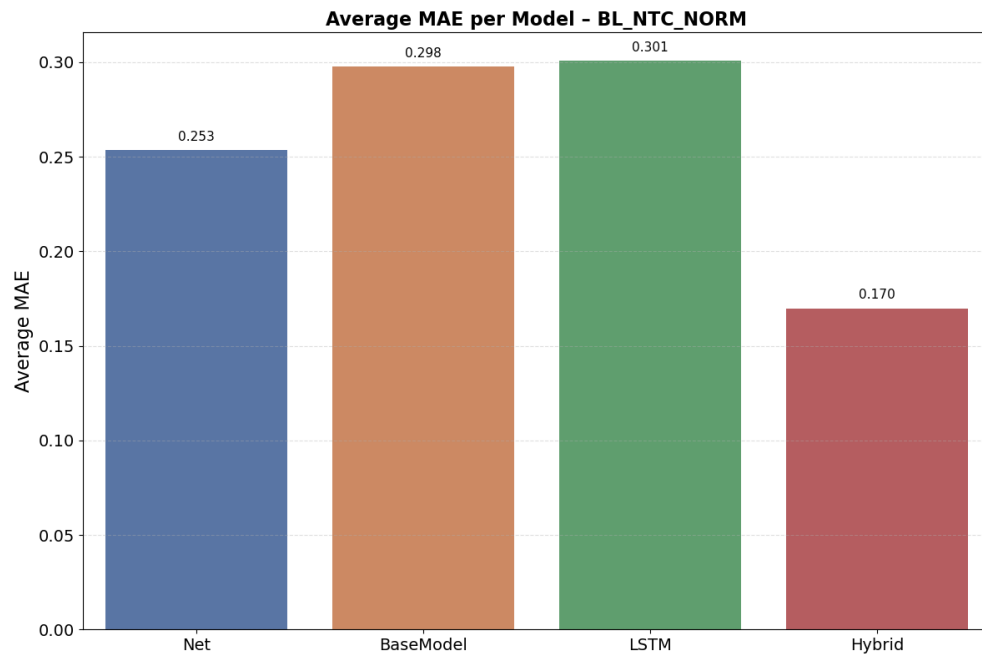


(c) LSTM MAE over epochs

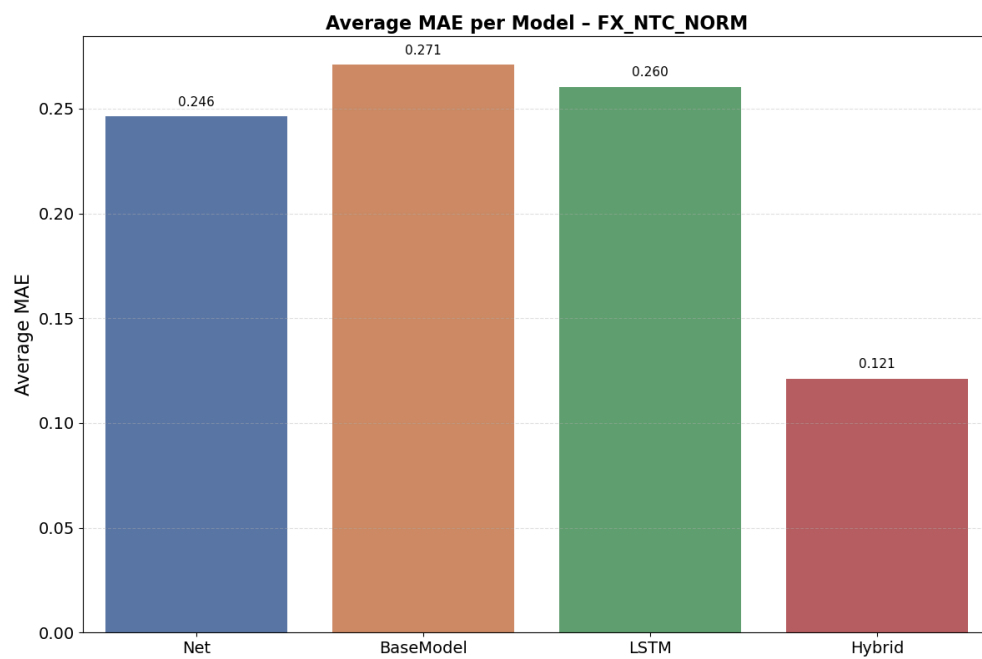


(d) Hybrid MAE over epochs

Figure A.3: Average training MAE (blue) and validation MAE (orange) over time for different models on NTC borders.

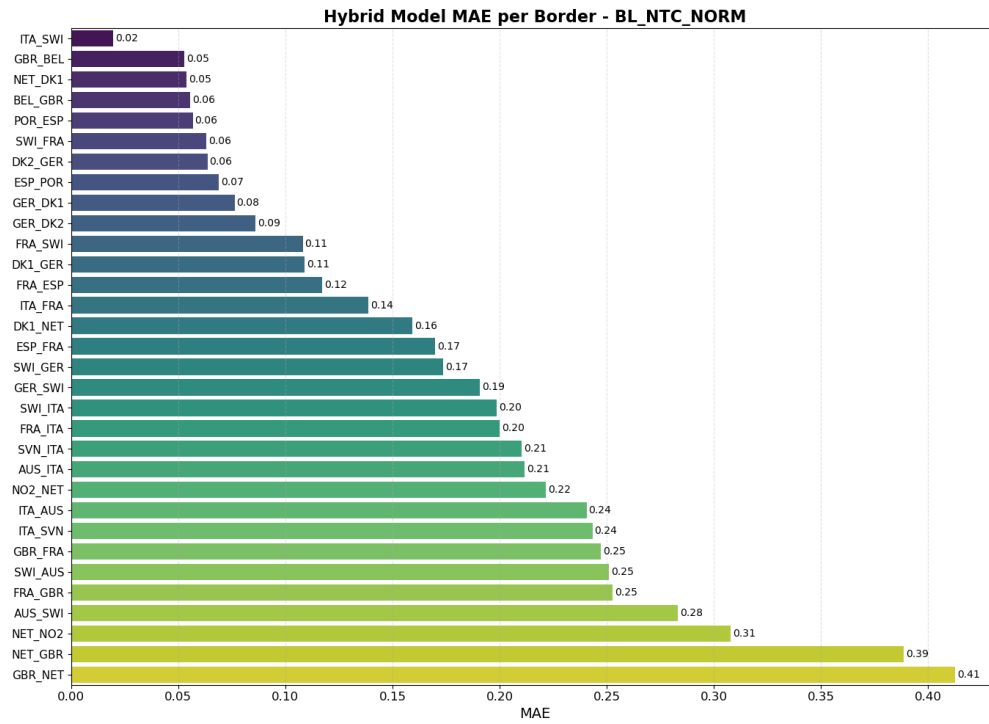


(a) Average MAE in testing on BL dataset

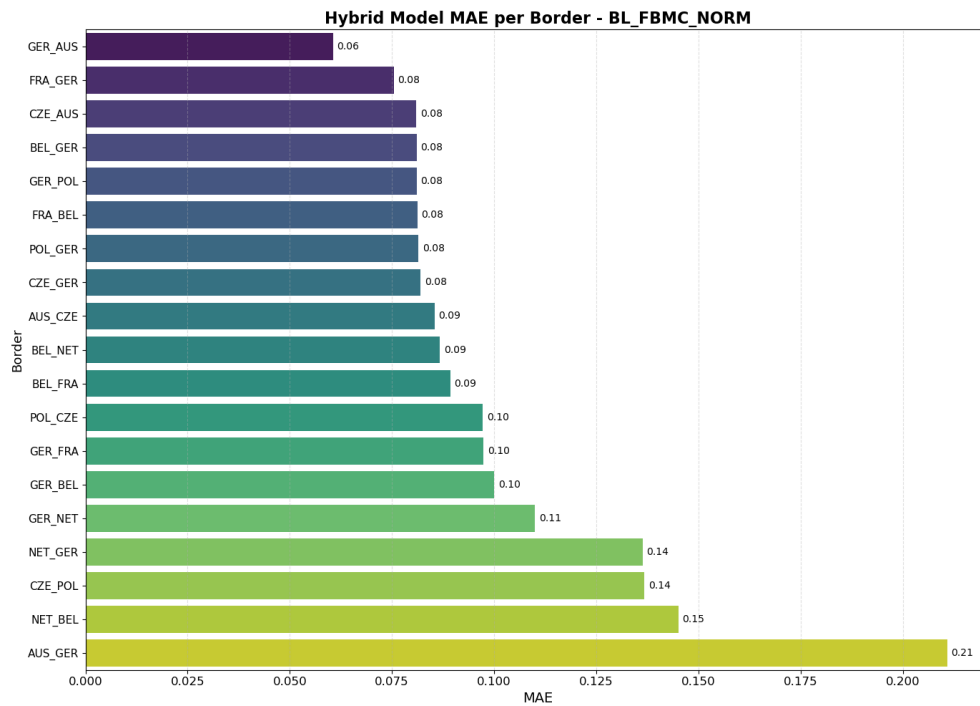


(b) Average MAE in Testing on FX dataset

Figure A.4: Average MAE in testing for each model on BL and FX dataset.



(a) Hybrid model MAE on NTC borders



(b) Hybrid model MAE on FBMC borders

Figure A.5: Overview of MAE on all borders for Hybrid model.